

REHATBIR SINGH

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EDUCATION

University of Wisconsin-Madison

BS Computer Science & Data Science (Dean's List) GPA: 4.0 / 4.0

Madison, WI, USA

Sep 2024 - May 2027

EXPERIENCE

Flux

Co-founder & Lead Software Engineer

Madison, WI, USA

Feb 2025 - Present

- Led product development for a personal app startup connecting student entrepreneurs by complementary skills, growing to 70+ MAU through rapid iteration and user feedback
- Made key decisions including adopting card-based profiles to enable future features and switching from Flutter to SwiftUI after validating iOS-dominant market to improve UX and development velocity
- Designed and implemented a semantic matching system using text embeddings, enabling flexible representations of users' skills and experiences

Al Masaood LLC

Software Developer Intern

Abu Dhabi, United Arab Emirates

May 2025 - Aug 2025

- Built and launched a React/SPFx timesheet tracker for 30+ technicians, saving 1,500+ sheets weekly through SAP-integrated SharePoint
- Assisted development of a legal chatbot using FastAPI, React, and RAG, saving the legal team 8 hrs/wk
- Partnered with the insurance department on an app for vehicle inventory and insurance, handling data modeling and SharePoint API integration

Data Science for Sustainable Development @ UW-Madison

Student Developer

Madison, WI, USA

Sep 2024 - Dec 2024

- Built a statistics dashboard for the open-source Red-CORAL crime visualization project, analyzing 1,200+ data points using React and Firebase
- Designed a modular filtering system using React Reducers, enabling flexible data slicing and real-time geospatial visualization

PROJECTS

MindMerge Rust, Tauri, React.js, Iroh, Fish Audio, Gemini API

<https://rehatsingh.cv/mindmerge>

Built a decentralized voice-controlled collaborative whiteboard for a 24h hackathon, implementing low-latency peer-to-peer state synchronization. Designed join-code-based swarm discovery using Gossip protocol. Integrated a voice-to-diagram pipeline generating structured SVG from natural language.

Dripr Swift, SwiftUI, Supabase, RealityKit

<https://rehatsingh.cv/coffee>

Developed a coffee brewing app featuring guided multi-phase timers powered by dynamic state machines using Combine. Designed a declarative recipe composer for a variety of brewers, and integrated RealityKit to render interactive 3D brewing tools in AR.

Blitz Board Python, JavaScript, Flask, PostgreSQL, Redis

<https://blitz.rehatsingh.cv>

Built a real-time multiplayer typing game using WebSockets and Redis-backed shared state to synchronize game lobbies across multiple Flask servers. Identified and debugged client desynchronization issues under concurrent updates. Containerized the system using Docker and self-hosted it using Nginx.

AWARDS

Honorable Mention in MadData Hackathon

dotData @ UW-Madison

Recognized for building & presenting Wavelength, an app that analyzes Spotify libraries to connect students with similar music tastes using data science techniques and social discovery algorithms.

Feb 2025

Winner of iOS App Dev Challenge

MDS UAE

Developed Earthify, an open-source sustainability marketplace app for recirculating used goods. Won first place overall and was a finalist in the environment category.

Dec 2021

SKILLS

Frontend: JavaScript/TypeScript, React.js, Tailwind, Swift, SwiftUI
Backend: Python, Flask, Django, Rust, Axum, Node.js, Next.js
Infrastructure: Docker, Nginx, Linux, PostgreSQL, MySQL, Firebase